

WEB ANALYTICS

BEGINNER-TO-JOB-READY ROADMAP

Learn it. Build it. Prove it. Publish it.

A robust printable checklist covering measurement strategy, GA4, Google Tag Manager, campaign tracking, data quality, dashboards, SQL, BigQuery, CRO, privacy, and portfolio development.

Name

Start date

Target role

PRINT • TICK • REVIEW MONTHLY • UPDATE YOUR PORTFOLIO

How to use this roadmap

Treat every checkbox as evidence-based progress, not passive study.

LEARN Understand the concept and terminology.	BUILD Implement it on a real or demo website.	PROVE Save screenshots, QA logs, queries, dashboards, and recommendations.
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The portfolio rule

A course certificate is supporting evidence. A working implementation, an analysis, and a clearly explained business recommendation are primary evidence.

Set your operating system

Done	Milestone / Task	Evidence
☐	Choose a target role: Web Analyst, Digital Analyst, Marketing Analyst, Analytics Implementation Specialist, or CRO Analyst.	Role written
☐	Create one main learning folder with subfolders: Notes, Tracking Plans, GTM, GA4, Dashboards, SQL, Portfolio.	Folder screenshot
☐	Create a portfolio home: Notion, personal website, GitHub Pages, or a clean PDF index.	URL/page
☐	Choose a practice website: your own site, a demo site, a local business site with permission, or a test ecommerce store.	Site selected
☐	Create a progress log and schedule two fixed study blocks per week.	Calendar
☐	Review the roadmap every four weeks and circle the next portfolio deliverable.	Review date

Recommended evidence labels

- ☐ KNOW = you can explain it without notes
- ☐ DO = you can perform it on a real property
- ☐ SHOW = you have portfolio-safe evidence
- ☐ TEACH = you can explain your decision and trade-offs

Personal goals and constraints

Roadmap at a glance

Progress from measurement thinking to technical implementation, analysis, and stakeholder communication.

#	Level	Core capability	Portfolio proof	Done
1	FOUNDATIONS	Business goals, customer journey, KPIs, metrics, dimensions, funnels, UTMs	Measurement plan	
2	IMPLEMENTATION	GA4, GTM, events, parameters, data layer, debugging, naming standards	Tagged practice site	
3	ANALYSIS	Acquisition, engagement, ecommerce, explorations, segmentation, spreadsheets	Insight case study	
4	COMMUNICATION	Looker Studio, data storytelling, stakeholder dashboards, recommendations	Executive dashboard	
5	ADVANCED	SQL, BigQuery, APIs, privacy, consent, experimentation, server-side concepts	Advanced portfolio project	

Baseline self-assessment

Tick	Capability	Score
☐	I can explain the difference between a business objective, KPI, metric, and dimension.	0 1 2 3
☐	I can design a measurement plan before opening an analytics tool.	0 1 2 3
☐	I can navigate GA4 reports and create useful explorations.	0 1 2 3
☐	I can configure and debug tags in Google Tag Manager.	0 1 2 3
☐	I can build a stakeholder-friendly dashboard.	0 1 2 3
☐	I can clean and analyse data in spreadsheets.	0 1 2 3
☐	I can write SQL queries for event-level analytics data.	0 1 2 3
☐	I can present findings as decisions, not just charts.	0 1 2 3

Scoring

0 = never done, 1 = familiar, 2 = can do with references, 3 = can do independently. Re-score every eight weeks.

PHASE 1 Business and Measurement Foundations

Learn to translate business questions into measurable user behaviours.

Learn

Done	Milestone / Task	Evidence
☐	Define business models: ecommerce, lead generation, content, SaaS, marketplace, nonprofit.	1-page notes
☐	Map the customer journey: awareness, consideration, conversion, retention, advocacy.	Journey map
☐	Separate objectives, strategies, tactics, KPIs, targets, metrics, and dimensions.	Definitions sheet
☐	Understand users, sessions, events, page/screen views, engagement, conversions/key events, revenue, retention.	Metric glossary
☐	Learn leading vs lagging indicators and input vs outcome metrics.	Examples
☐	Learn rates and denominators: conversion rate, engagement rate, CTR, completion rate, revenue per user.	Formula sheet
☐	Understand funnels, cohorts, segments, attribution, channel grouping, and campaign taxonomy.	Concept map
☐	Practise asking: What decision will this metric change?	10 questions

Practise

Done	Milestone / Task	Evidence
☐	Choose one business and write its top three business objectives.	Document
☐	For each objective, define one KPI, target, segment, and decision owner.	KPI tree
☐	Create a customer journey with measurable actions at each stage.	Journey diagram
☐	Write a KPI dictionary with definition, formula, scope, owner, source, and caveat.	Dictionary
☐	Design a funnel and identify where instrumentation is required.	Funnel map

Portfolio milestone

Publish a Measurement Strategy Pack: business context, KPI tree, customer journey, funnel, KPI dictionary, and measurement risks.

Exit criteria

- ☐ I can reject vanity metrics and explain why.
- ☐ I can define a KPI with a formula and decision owner.
- ☐ I can map a user journey to measurable events.
- ☐ I can explain metric scope and denominator.

PHASE 2 Campaign Tracking and Acquisition Analytics

Build a clean campaign taxonomy before analysing marketing performance.

Learn

Done	Milestone / Task	Evidence
☐	Understand source, medium, campaign, content, term, referrer, direct traffic, and channel groups.	Glossary
☐	Learn UTM parameters and when not to use UTMs, especially internal links.	Rules sheet
☐	Create naming standards for paid social, email, influencer, affiliate, QR, and partner links.	Naming guide
☐	Understand landing pages, first-user acquisition, session acquisition, assisted journeys, and attribution limitations.	Comparison notes
☐	Learn basic paid media metrics: impressions, clicks, CTR, CPC, CPM, conversions, CPA, ROAS.	Formula sheet
☐	Understand cross-domain and self-referral problems conceptually.	Issue notes

Practise

Done	Milestone / Task	Evidence
☐	Build a UTM generator spreadsheet with validation lists.	Spreadsheet
☐	Create 20 correctly tagged campaign URLs for different channels.	URL sheet
☐	Test links and verify parameters persist through redirects.	QA log
☐	Create a campaign naming policy with good and bad examples.	Policy PDF
☐	Analyse acquisition performance using the GA4 demo account.	Analysis

Portfolio milestone

Publish a Campaign Measurement Kit: UTM builder, taxonomy, QA checklist, acquisition analysis, and three optimisation recommendations.

Exit criteria

- ☐ I can build consistent campaign URLs.
- ☐ I can diagnose common attribution hygiene issues.
- ☐ I can compare channels using business-relevant outcomes.
- ☐ I can explain why UTMs should not be added to internal links.

PHASE 3 Google Analytics 4 Core

Become fluent in the interface, event model, reports, and explorations.

Learn

Done	Milestone / Task	Evidence
☐	Set up account, property, data stream, enhanced measurement, internal traffic, and referral exclusions conceptually.	Setup notes
☐	Understand the GA4 event model: event names, parameters, user properties, item arrays, and scope.	Event model diagram
☐	Learn automatically collected, enhanced measurement, recommended, and custom events.	Event table
☐	Understand standard reports, realtime, DebugView, advertising reports, and explorations.	Navigation map
☐	Learn free-form, funnel, path, cohort, segment overlap, and user exploration use cases.	Use-case sheet
☐	Understand custom dimensions and metrics, key events, audiences, and data retention.	Admin checklist
☐	Learn common reporting differences, thresholds, sampling concepts, cardinality, and data freshness caveats.	Caveat log

Practise

Done	Milestone / Task	Evidence
☐	Access the official GA4 demo account and save five report observations.	Screenshots
☐	Create a free-form exploration comparing two meaningful segments.	Exploration
☐	Build a funnel exploration and identify the largest drop-off.	Funnel
☐	Build a path exploration from a conversion or exit point.	Path
☐	Create a custom report or collection outline for a stakeholder.	Design
☐	Write five recommendations with impact, confidence, effort, and evidence.	Recommendation table

Portfolio milestone

Publish a GA4 Behaviour Analysis: business question, method, exploration screenshots, findings, recommendations, limitations, and appendix.

Exit criteria

- ☐ I can choose the correct report or exploration for a question.
- ☐ I can create and compare segments.
- ☐ I can explain event/parameter scope.
- ☐ I can turn findings into prioritised actions.

PHASE 4 Google Tag Manager and the Data Layer

Implement analytics safely, consistently, and with version-controlled QA evidence.

Learn

Done	Milestone / Task	Evidence
☐	Understand containers, workspaces, tags, triggers, variables, folders, versions, and publishing.	Architecture notes
☐	Install and validate a web container on a test site.	Container proof
☐	Learn click, form, visibility, scroll, history change, custom event, and timer triggers.	Trigger matrix
☐	Learn built-in variables, data layer variables, lookup tables, regex tables, and custom JavaScript risks.	Variable guide
☐	Understand dataLayer pushes, event-driven tagging, ecommerce objects, and naming conventions.	Data layer schema
☐	Learn GA4 configuration/Google tag concepts and GA4 event tags.	Tag map
☐	Use Preview mode and Tag Assistant to inspect firing order, parameters, consent state, and errors.	Debug notes
☐	Learn version naming, change notes, rollback, environments, and least-privilege access.	Governance checklist

Practise

Done	Milestone / Task	Evidence
☐	Create a tracking plan before building tags.	Tracking plan
☐	Track outbound clicks, file downloads, form success, CTA clicks, and video interactions.	Tag set
☐	Implement at least one custom dataLayer event.	Code + proof
☐	Create an ecommerce or lead-generation event sequence.	Event map
☐	Perform happy-path, failure-path, duplicate-event, and edge-case testing.	QA log
☐	Publish with a clear version name and rollback note.	Version screenshot

Portfolio milestone

Publish an Implementation Case Study: tracking plan, event schema, GTM screenshots, dataLayer examples, test matrix, issues found, and final architecture.

Exit criteria

- ☐ I can implement from a tracking plan rather than ad hoc.
- ☐ I can debug why a tag did or did not fire.
- ☐ I can prevent duplicate events.
- ☐ I can document and safely publish a container version.

PHASE 5 Data Quality, Debugging, and Governance

Trustworthy analysis starts with trustworthy collection.

Learn

Done	Milestone / Task	Evidence
☐	Learn an analytics QA lifecycle: requirements, build, test, release, monitor, document.	QA lifecycle
☐	Understand duplicate events, missing parameters, inconsistent naming, PII risk, bot/internal traffic, and broken referrals.	Issue catalog
☐	Use browser developer tools: console, network requests, cookies/storage, and page source.	DevTools notes
☐	Understand event validation, schema checks, expected counts, reconciliation, and anomaly detection.	Validation plan
☐	Create naming conventions and ownership rules for accounts, properties, containers, events, parameters, reports, and dashboards.	Governance guide
☐	Learn change management, access reviews, version logs, rollback procedures, and documentation standards.	Control checklist

Practise

Done	Milestone / Task	Evidence
☐	Audit 20 events against a tracking plan.	Audit table
☐	Create a test matrix for desktop/mobile, browser, consent state, and user state.	Test matrix
☐	Document five intentionally created tagging defects and fixes.	Bug log
☐	Create an analytics release checklist.	Release checklist
☐	Define monthly data-quality monitoring metrics.	Monitoring spec

Portfolio milestone

Publish an Analytics QA Audit with severity ratings, reproduction steps, evidence, recommended fixes, owners, and re-test status.

Exit criteria

- ☐ I can verify data before using it.
- ☐ I can write reproducible bug reports.
- ☐ I can distinguish collection defects from reporting differences.
- ☐ I can design lightweight analytics governance.

PHASE 6 Spreadsheets and Analytical Thinking

Use simple tools rigorously before reaching for advanced platforms.

Learn

Done	Milestone / Task	Evidence
☐	Master sorting, filtering, tables, named ranges, validation, conditional formatting, and pivot tables.	Skills sheet
☐	Learn formulas: IF/IFS, SUMIFS, COUNTIFS, XLOOKUP/INDEX-MATCH, text functions, date functions, error handling.	Formula exercises
☐	Calculate rates, weighted averages, percent change, contribution, index values, and confidence-aware comparisons.	Calculation sheet
☐	Clean campaign names, URLs, dates, blanks, duplicates, and inconsistent categories.	Cleaning exercises
☐	Build repeatable analysis templates and separate raw, clean, calculation, and output tabs.	Workbook structure
☐	Learn basic descriptive statistics and avoid misleading averages.	Statistics notes

Practise

Done	Milestone / Task	Evidence
☐	Clean a messy web analytics export.	Before/after
☐	Build a channel performance pivot with calculated KPIs.	Pivot
☐	Create a landing-page scorecard.	Scorecard
☐	Perform a week-over-week and year-over-year comparison.	Trend analysis
☐	Write an executive summary with three findings and three actions.	Summary

Portfolio milestone

Publish a Spreadsheet Analysis Pack with raw data, cleaning log, formulas, pivot tables, charts, findings, and documentation.

Exit criteria

- ☐ I can make an analysis reproducible.
- ☐ I can check denominators and totals.
- ☐ I can identify outliers and data-quality issues.
- ☐ I can communicate the decision, not only the calculation.

PHASE 7 Looker Studio and Data Storytelling

Build dashboards that answer stakeholder questions quickly and accurately.

Learn

Done	Milestone / Task	Evidence
☐	Understand data sources, connectors, dimensions, metrics, aggregation, filters, date ranges, and controls.	Concept notes
☐	Learn scorecards, time series, tables, bar charts, scatter plots, funnels, and when not to use each.	Chart guide
☐	Create calculated fields, CASE statements, blended data concepts, parameters, and report-level controls.	Calculation sheet
☐	Learn dashboard hierarchy: context, KPI status, drivers, segments, diagnostic detail, actions.	Wireframe
☐	Apply visual design principles: alignment, spacing, consistency, accessible contrast, number formatting, and annotation.	Design checklist
☐	Design for executives, channel managers, product teams, and analysts differently.	Audience matrix

Practise

Done	Milestone / Task	Evidence
☐	Sketch a dashboard before connecting data.	Wireframe
☐	Build an executive one-page dashboard.	Dashboard
☐	Build a diagnostic second page with segment and date controls.	Dashboard
☐	Add metric definitions and data freshness notes.	Info panel
☐	Conduct a five-minute usability test with another person.	Feedback log
☐	Export a clean PDF snapshot for a portfolio case study.	PDF

Portfolio milestone

Publish a Stakeholder Dashboard Case Study: audience, questions, wireframe, calculations, final dashboard, usability changes, and action summary.

Exit criteria

- ☐ Each chart answers a named question.
- ☐ Every KPI has a definition and comparison.
- ☐ The dashboard separates status from diagnosis.
- ☐ I can explain aggregation and filter behaviour.

PHASE 8 CRO, Experimentation, and Qualitative Analytics

Combine behavioural data with hypotheses and user evidence.

Learn

Done	Milestone / Task	Evidence
☐	Learn conversion research: heuristic review, funnel analysis, segmentation, surveys, recordings, heatmaps, and user testing.	Research map
☐	Understand hypotheses: observation, evidence, proposed change, expected outcome, and primary metric.	Hypothesis template
☐	Learn experiment basics: control, variant, randomisation, sample size concepts, duration, primary/guardrail metrics, and novelty effects.	Experiment notes
☐	Understand statistical significance vs practical significance at a conceptual level.	Concept sheet
☐	Learn prioritisation frameworks such as impact-confidence-effort and PIE/RICE concepts.	Prioritisation sheet
☐	Understand why pre/post comparisons alone are weak evidence.	Caveat notes

Practise

Done	Milestone / Task	Evidence
☐	Audit a landing page using analytics and heuristic evidence.	Audit
☐	Create five testable hypotheses.	Backlog
☐	Prioritise hypotheses with a transparent scoring method.	Scorecard
☐	Define primary, secondary, and guardrail metrics for one test.	Metric plan
☐	Create a mock experiment readout with decision rules.	Readout

Portfolio milestone

Publish a CRO Research Case Study: evidence, funnel, friction points, hypotheses, prioritisation, experiment design, and expected learning.

Exit criteria

- ☐ I can distinguish a finding from a hypothesis.
- ☐ I can define a testable change and metric.
- ☐ I can identify confounders and limitations.
- ☐ I can prioritise ideas transparently.

PHASE 9 SQL and BigQuery for Analytics

Move from interface reports to event-level, reproducible analysis.

Learn

Done	Milestone / Task	Evidence
☐	Learn database concepts: tables, rows, columns, schemas, keys, nulls, data types, and grain.	Database notes
☐	Write SELECT, WHERE, CASE, GROUP BY, HAVING, ORDER BY, LIMIT, and basic functions.	Query set
☐	Learn JOINS, CTEs, subqueries, window functions, date functions, and safe division.	Advanced query set
☐	Understand nested and repeated event data, UNNEST concepts, event parameters, user/session identifiers, and ecommerce items.	GA4 schema map
☐	Learn query cost awareness, partition filters, reusable views, and documentation.	Cost checklist
☐	Reconcile SQL outputs with interface reports and document expected differences.	Reconciliation notes

Practise

Done	Milestone / Task	Evidence
☐	Complete at least 40 SQL exercises.	Exercise log
☐	Query sessions/users/events by date and channel.	SQL file
☐	Build a funnel from event-level data.	SQL file
☐	Calculate landing-page conversion rate with a clearly defined denominator.	SQL file
☐	Create a cohort or retention query.	SQL file
☐	Create a documented dataset or view for a dashboard.	View spec

Portfolio milestone

Publish a BigQuery/SQL Case Study with business question, schema notes, commented queries, validation, results, chart, recommendations, and cost considerations.

Exit criteria

- ☐ I can identify the grain of a dataset.
- ☐ I can avoid double-counting after joins/unnesting.
- ☐ I can write readable CTE-based queries.
- ☐ I can validate SQL results against expectations.

PHASE 10 Privacy, Consent, and Responsible Analytics

Design measurement that minimises risk and respects user choices.

Learn

Done	Milestone / Task	Evidence
☐	Understand data minimisation, purpose limitation, retention, access control, and documentation principles.	Principles sheet
☐	Identify personal data and prohibited collection risks, including URLs, form fields, and free text.	Risk inventory
☐	Understand consent-management concepts and tag behaviour before/after consent.	Consent flow
☐	Learn Google consent mode concepts and how to verify consent signals in Tag Assistant.	Verification notes
☐	Understand cookie/storage basics, cross-domain implications, and vendor governance.	Storage notes
☐	Learn role-based access, audit logs, account ownership, offboarding, and incident response basics.	Access checklist
☐	Document assumptions and consult the relevant legal/privacy specialist for jurisdiction-specific requirements.	Escalation rule

Practise

Done	Milestone / Task	Evidence
☐	Create a data inventory for every event and parameter on a practice site.	Inventory
☐	Perform a PII-risk review of page URLs, query strings, forms, and dataLayer values.	Risk audit
☐	Create consent-state QA scenarios and expected tag behaviour.	Test matrix
☐	Write an analytics retention and access review checklist.	Policy checklist
☐	Add a privacy and limitations section to every portfolio project.	Updated cases

Portfolio milestone

Publish a Privacy-Aware Measurement Review: data inventory, risk findings, consent QA, remediation plan, and governance checklist. Do not present it as legal advice.

Exit criteria

- ☐ I can spot common PII collection risks.
- ☐ I can test tag behaviour across consent states.
- ☐ I can apply least privilege and access reviews.
- ☐ I can explain when specialist legal guidance is required.

PHASE 11 Advanced and Optional Specialisations

Choose depth based on the role you want, not because every tool is mandatory.

Learn

Done	Milestone / Task	Evidence
☐	Analytics engineering: transformation layers, data models, dbt concepts, testing, documentation.	Track chosen
☐	APIs and automation: GA4 Data API, reporting automation, scheduled extracts, alerting.	Track chosen
☐	Server-side measurement: server containers, first-party endpoints, governance, cost, and limitations.	Track chosen
☐	Product analytics: activation, retention, feature adoption, lifecycle analysis, and experimentation.	Track chosen
☐	Ecommerce analytics: item-level data, merchandising, checkout, refunds, promotions, and LTV.	Track chosen
☐	B2B/lead analytics: CRM integration, lead quality, offline outcomes, and pipeline attribution.	Track chosen
☐	Technical web fundamentals: HTML, CSS selectors, DOM, JavaScript events, network requests, cookies, and APIs.	Track chosen

Practise

Done	Milestone / Task	Evidence
☐	Select one specialisation and define a 6-week mini-roadmap.	Plan
☐	Build one advanced proof-of-concept.	Prototype
☐	Document architecture, risks, cost, and alternatives.	Design note
☐	Explain when the advanced solution is unnecessary.	Trade-off note

Portfolio milestone

Publish one specialist project only after the core roadmap is strong. A narrow, well-explained project is better than a shallow list of advanced tools.

Exit criteria

- ☐ I chose a specialisation aligned with target jobs.
- ☐ I can explain trade-offs and operational cost.
- ☐ I can document architecture clearly.
- ☐ The project solves a real measurement problem.

Portfolio Architecture

Make every project easy to understand in under two minutes, then easy to inspect in depth.

Your portfolio is not a screenshot gallery

It should show how you define the problem, design measurement, verify data, analyse evidence, make a recommendation, and acknowledge limitations.

Required structure for every case study

Done	Milestone / Task	Evidence
☐	1. Project title and one-sentence outcome.	Top section
☐	2. Business context and stakeholder question.	Context
☐	3. Scope, assumptions, permissions, and privacy safeguards.	Scope
☐	4. Measurement plan or analytical method.	Method
☐	5. Implementation or data preparation.	Build
☐	6. QA and validation evidence.	Proof
☐	7. Analysis with clearly defined metrics and segments.	Analysis
☐	8. Findings separated from interpretations.	Findings
☐	9. Prioritised recommendations with impact, confidence, and effort.	Actions
☐	10. Limitations, risks, and what you would do next.	Reflection
☐	11. Supporting assets: dashboard, SQL, spreadsheet, schema, screenshots, README.	Appendix
☐	12. Three resume bullets with action, tool, method, and result/expected impact.	Resume

Recommended portfolio formats

- ☐ Portfolio home page with 4-6 strong projects, not 20 weak projects.
- ☐ One clean case-study page per project with a PDF export.
- ☐ GitHub repository for SQL, schemas, sample data, and README files.
- ☐ Looker Studio link or screenshots using demo/synthetic data.
- ☐ Redacted screenshots for GTM/GA4; never expose client IDs, emails, or private data.
- ☐ A short 2-4 minute walkthrough video for your strongest project.

Portfolio quality scorecard

Tick	Criterion	Score
☐	Problem is specific and business-relevant.	/5
☐	Method is appropriate and reproducible.	/5
☐	Data quality is tested and documented.	/5

Tick	Criterion	Score
☐	Analysis uses correct definitions and denominators.	/5
☐	Recommendations are prioritised and actionable.	/5
☐	Visuals are clear and labelled.	/5
☐	Limitations and privacy are addressed.	/5
☐	Assets are organised and easy to inspect.	/5
☐	Writing is concise and understandable.	/5
☐	You can defend every decision in an interview.	/5

Publishing threshold

Do not publish below 38/50. Improve the weakest section before adding another project.

PROJECT 1 Measurement Strategy Pack

Portfolio Project 1 | Beginner

Project brief

Choose a real or fictional business and create the measurement foundation before any tool implementation.

Build checklist

Done	What to build	Proof
☐	Define the business model, audience, and top three objectives.	Brief
☐	Build a KPI tree linking objectives to user behaviours.	KPI tree
☐	Map the customer journey and funnel.	Diagram
☐	Create a KPI dictionary with formulas and owners.	Dictionary
☐	Create an event and parameter tracking plan.	Tracking plan
☐	List data risks, assumptions, and exclusions.	Risk log

Questions your analysis must answer

- ☐ Which business decisions will this measurement system support?
- ☐ Which KPIs are leading versus lagging?
- ☐ Where could the metric be gamed or misunderstood?
- ☐ Which events are essential, optional, or unnecessary?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Case-study page or PDF.	Link
☐	KPI dictionary spreadsheet.	File
☐	Journey/funnel diagram.	Image
☐	Tracking plan.	File
☐	Three resume bullets.	Text

Definition of done

A reviewer can understand what should be measured, why it matters, how it will be defined, and what risks remain.

PROJECT 2 GA4 + GTM Implementation

Portfolio Project 2 | Core

Project brief

Instrument a practice site using a documented event schema and a controlled GTM release process.

Build checklist

Done	What to build	Proof
☐	Install and validate GTM/Google tag on a test site.	Proof
☐	Implement at least five meaningful events.	GTM
☐	Use event parameters and at least one dataLayer event.	Schema
☐	Create a test plan covering consent and device/browser states.	QA
☐	Validate in Preview/Tag Assistant and GA4 DebugView.	Screenshots
☐	Publish with version notes and rollback instructions.	Version

Questions your analysis must answer

- ☐ Does every event map to a business question?
- ☐ Could any event fire twice or miss a valid interaction?
- ☐ Are names and parameters consistent?
- ☐ What happens when consent is denied or changes?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Architecture diagram.	Image
☐	Tracking plan and event schema.	File
☐	Redacted GTM screenshots.	Images
☐	QA test matrix and bug log.	File
☐	Version/change log.	File
☐	Case-study narrative.	Link

Definition of done

The implementation is testable, reproducible, privacy-aware, and understandable without access to the live container.

PROJECT 3 Campaign Taxonomy and Acquisition Analysis

Portfolio Project 3 | Core

Project brief

Design a campaign naming system and analyse channel performance using GA4 demo or synthetic data.

Build checklist

Done	What to build	Proof
☐	Create UTM naming rules and controlled vocabulary.	Guide
☐	Build a UTM generator spreadsheet.	Sheet
☐	Generate and QA sample links.	Log
☐	Create an acquisition performance table.	Analysis
☐	Segment by campaign/landing page/device where relevant.	Analysis
☐	Recommend budget or landing-page actions.	Actions

Questions your analysis must answer

- ☐ Which campaigns bring volume versus quality?
- ☐ Are conclusions based on rates and absolute counts?
- ☐ Could inconsistent tagging distort the result?
- ☐ What decision would change based on this analysis?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Taxonomy guide.	PDF
☐	UTM builder.	Sheet
☐	QA checklist.	File
☐	Analysis workbook/dashboard.	Link
☐	Recommendation summary.	PDF

Definition of done

The system prevents naming chaos and the analysis leads to clear acquisition decisions.

PROJECT 4 GA4 Funnel and Journey Analysis

Portfolio Project 4 | Core

Project brief

Use explorations to identify a meaningful journey problem and propose evidence-based actions.

Build checklist

Done	What to build	Proof
☐	Define the conversion journey and step criteria.	Definition
☐	Create a funnel exploration.	Screenshot
☐	Compare at least two useful segments.	Screenshot
☐	Create a supporting path or free-form exploration.	Screenshot
☐	Validate metric definitions and time range.	QA
☐	Prioritise three actions using impact-confidence-effort.	Actions

Questions your analysis must answer

- ☐ Where is the largest material drop-off?
- ☐ Is the drop-off consistent across devices or channels?
- ☐ What alternative explanations exist?
- ☐ Which recommendation can be tested first?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Case-study PDF.	PDF
☐	Exploration screenshots with annotations.	Images
☐	Metric definition appendix.	File
☐	Recommendation backlog.	Sheet
☐	Two-minute walkthrough.	Video

Definition of done

The case study separates observed behaviour from causal claims and proposes a testable next step.

PROJECT 5 Stakeholder Dashboard

Portfolio Project 5 | Core

Project brief

Build a dashboard for a named audience and decision cadence.

Build checklist

Done	What to build	Proof
☐	Write the audience, decisions, and refresh cadence.	Brief
☐	Sketch a wireframe before building.	Wireframe
☐	Create KPI status, trend, driver, and diagnostic sections.	Dashboard
☐	Add filters, date controls, definitions, and freshness notes.	Controls
☐	Test calculations and filter interactions.	QA
☐	Run a usability test and document changes.	Feedback

Questions your analysis must answer

- ☐ Can the stakeholder see what changed in 30 seconds?
- ☐ Can they identify why it changed in two minutes?
- ☐ Does each visual answer one question?
- ☐ Are comparisons, units, and targets clear?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Dashboard link or PDF snapshot.	Link
☐	Wireframe.	Image
☐	Metric dictionary.	File
☐	Calculated field documentation.	File
☐	Usability feedback log.	File

Definition of done

A non-analyst can identify status, drivers, and required action without your explanation.

PROJECT 6 SQL / BigQuery Event Analysis

Portfolio Project 6 | Advanced

Project brief

Answer a business question using event-level analytics data and reproducible SQL.

Build checklist

Done	What to build	Proof
☐	Define dataset grain and relevant fields.	Schema
☐	Write clean CTE-based SQL with comments.	SQL
☐	Handle nested/repeated fields carefully.	SQL
☐	Validate totals and denominators.	QA
☐	Create one useful chart or table from results.	Output
☐	Document cost/performance considerations.	Notes

Questions your analysis must answer

- ☐ What is the grain before and after each join or UNNEST?
- ☐ Could rows be duplicated?
- ☐ How does this result differ from the interface and why?
- ☐ What business action follows from the query?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	README with question and method.	GitHub
☐	SQL files.	GitHub
☐	Schema/data dictionary.	File
☐	Validation checks.	File
☐	Case-study chart and recommendations.	PDF

Definition of done

Another analyst can run the query, understand the grain, reproduce the result, and challenge your assumptions.

PROJECT 7 Analytics QA and Governance Audit

Portfolio Project 7 | Core

Project brief

Audit a practice implementation, identify defects, and design a sustainable control process.

Build checklist

Done	What to build	Proof
☐	Create an audit scope and event inventory.	Scope
☐	Test events against requirements.	Matrix
☐	Record defects with severity and reproduction steps.	Bug log
☐	Recommend fixes and ownership.	Plan
☐	Re-test fixes and record status.	Evidence
☐	Create release and monthly monitoring checklists.	Controls

Questions your analysis must answer

- ☐ Which defects materially change decisions?
- ☐ Which issues are implementation, configuration, or interpretation problems?
- ☐ How will regressions be detected?
- ☐ Who owns each control?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Audit report.	PDF
☐	Test matrix.	Sheet
☐	Bug log.	Sheet
☐	Governance checklist.	File
☐	Before/after evidence.	Images

Definition of done

The audit is prioritised, reproducible, and includes controls that prevent the same defects returning.

PROJECT 8 CRO Research and Experiment Plan

Portfolio Project 8 | Advanced

Project brief

Combine quantitative and qualitative evidence to design a defensible experiment.

Build checklist

Done	What to build	Proof
☐	Select a page/journey and define the conversion objective.	Brief
☐	Analyse funnel or segment behaviour.	Analysis
☐	Perform a heuristic and qualitative review.	Audit
☐	Create at least five hypotheses.	Backlog
☐	Prioritise with a transparent framework.	Scorecard
☐	Design one experiment with primary and guardrail metrics.	Plan

Questions your analysis must answer

- ☐ What evidence supports the hypothesis?
- ☐ What user problem is being solved?
- ☐ What result would support, reject, or complicate the hypothesis?
- ☐ What risks or unintended effects require guardrails?

Portfolio publishing checklist

Done	Portfolio asset	Location
☐	Research summary.	PDF
☐	Annotated page audit.	Images
☐	Hypothesis backlog.	Sheet
☐	Experiment plan/readout template.	File
☐	Limitations and next steps.	Text

Definition of done

The proposed experiment is grounded in evidence, measurable, and explicit about uncertainty.

12-Week Intensive Plan

For learners studying approximately 8-12 focused hours per week.

Week	Focus	Required output	Done
1	Business foundations, KPI tree, customer journey	Measurement strategy draft	☐
2	Metrics, funnels, attribution, campaign taxonomy	KPI dictionary + UTM guide	☐
3	GA4 reports, realtime, admin concepts	Demo-account report notes	☐
4	GA4 explorations and segmentation	Funnel/journey case draft	☐
5	GTM tags, triggers, variables	Basic implementation	☐
6	Data layer, custom events, debugging	Implementation + QA evidence	☐
7	Data-quality audit and governance	Audit report	☐
8	Spreadsheet analysis	Analysis workbook	☐
9	Looker Studio design and build	Dashboard v1	☐
10	SQL foundations	20-30 queries	☐
11	BigQuery/event analysis or CRO specialisation	Advanced project draft	☐
12	Portfolio polishing, walkthrough, interview practice	4 publishable projects	☐

Do not rush the portfolio

If a week slips, move the date. Do not mark a skill complete without implementation or analysis evidence.

Weekly review notes

24-Week Steady Plan

For learners studying approximately 4-6 focused hours per week.

Learning blocks

Done	Focus block	Output
☐	Weeks 1-2: Business models, journeys, KPI trees	Measurement Strategy Pack
☐	Weeks 3-4: Metrics, funnels, UTMs, acquisition	Campaign Measurement Kit
☐	Weeks 5-7: GA4 reports, admin concepts, explorations	GA4 analysis case
☐	Weeks 8-11: GTM, data layer, implementation, debugging	Implementation case
☐	Weeks 12-13: QA, governance, privacy basics	QA audit
☐	Weeks 14-15: Spreadsheet cleaning and analysis	Workbook case
☐	Weeks 16-18: Looker Studio and storytelling	Dashboard case
☐	Weeks 19-21: SQL foundations and BigQuery concepts	SQL case
☐	Weeks 22-23: CRO/experimentation or another specialisation	Specialist case
☐	Week 24: Portfolio edit, mock interview, applications	Portfolio launch

Dates and adjustments

Job-Readiness Checklist

Use this before applying for web analytics roles.

Core capability evidence

Done	Milestone / Task	Evidence
☐	I have at least four strong, published case studies.	Links
☐	I can create a measurement plan from a business brief.	Project
☐	I can analyse GA4 reports and explorations.	Project
☐	I can implement and debug common GTM events.	Project
☐	I can create and defend a dashboard design.	Project
☐	I can clean and analyse exports in spreadsheets.	Project
☐	I can explain attribution and campaign-tagging limitations.	Interview
☐	I can conduct analytics QA and document defects.	Project
☐	I understand privacy, consent, and PII risks.	Project
☐	I can write basic-to-intermediate SQL or have a clear plan to learn it.	Queries
☐	I can present three findings in plain business language.	Presentation
☐	I can explain the limitations of my analysis.	Interview

Application assets

Done	Milestone / Task	Evidence
☐	One-page resume tailored to the role.	Ready
☐	LinkedIn headline and About section aligned with analytics.	Ready
☐	Portfolio URL visible in resume and LinkedIn.	Ready
☐	GitHub or file repository is organised and documented.	Ready
☐	Two-minute introduction linking background to analytics.	Ready
☐	Six STAR stories: debugging, stakeholder, ambiguity, failure, improvement, insight.	Ready
☐	Ten target companies and role keywords tracked.	Ready
☐	Application tracker with date, role, status, follow-up, and feedback.	Ready

Resume Evidence Builder

Turn portfolio proof into defensible resume bullets and interview talking points.

Build each bullet from evidence

Done	Milestone / Task	Evidence
☐	Choose one portfolio project and state the business problem.	Project
☐	Name the tool and the specific analytical or implementation method.	Method
☐	Describe the validation or QA performed.	Proof
☐	State the decision, recommendation, or measurable outcome.	Outcome
☐	Quantify only what can be supported honestly.	Checked
☐	Tailor terminology to the target job description without keyword stuffing.	Tailored

Action + tool + method + decision/result

Example: Built a GA4/GTM measurement plan and five-event implementation for a demo lead-generation site; validated firing and parameters across device and consent scenarios, then documented defects and remediation priorities.

Draft three resume bullets / talking points

Interview Practice Bank

Answer each aloud, then improve the answer using evidence from your portfolio.

Questions

Done	Practice question	Score
☐	How would you measure success for a lead-generation website?	1 2 3
☐	What is the difference between a metric and a dimension?	1 2 3
☐	How would you design a tracking plan for a new feature?	1 2 3
☐	An event count doubled overnight. How would you investigate?	1 2 3
☐	Why might GA4 reports and explorations show different numbers?	1 2 3
☐	How would you diagnose a sudden rise in Direct traffic?	1 2 3
☐	When should UTMs be used, and when should they not be used?	1 2 3
☐	How would you validate a GTM implementation before publishing?	1 2 3
☐	What is the purpose of a data layer?	1 2 3
☐	How would you choose the steps in a conversion funnel?	1 2 3
☐	A mobile conversion rate is lower than desktop. What would you do next?	1 2 3
☐	What makes a dashboard useful rather than decorative?	1 2 3
☐	How would you explain a decline in conversion to a non-technical stakeholder?	1 2 3
☐	How do you avoid double-counting in SQL?	1 2 3
☐	How would you prioritise analytics defects?	1 2 3
☐	What privacy risks do analytics implementations create?	1 2 3
☐	What is one portfolio project you would redesign and why?	1 2 3
☐	Tell me about a time you found a data-quality issue.	1 2 3
☐	How do you distinguish correlation from causation?	1 2 3
☐	What would you measure during an A/B test?	1 2 3

Answer standard

Give the business context, define the metric, describe the method, explain validation, state the decision, and name limitations.

Monthly Progress Review

Print or duplicate this page for each month.

Month / dates	_____
Hours studied	_____
Phases worked on	_____
Portfolio output completed	_____
Strongest new skill	_____
Biggest data/technical mistake	_____
How I fixed or documented it	_____
One concept I can now teach	_____
One weak area to revisit	_____
Next month's primary deliverable	_____

Portfolio maintenance

Done	Milestone / Task	Evidence
☐	Update project status and links.	Done
☐	Replace weak screenshots with annotated evidence.	Done
☐	Improve one README or case-study explanation.	Done
☐	Check that no confidential or personal data is visible.	Done
☐	Add one new reflection or limitation.	Done
☐	Archive old drafts and keep a clean public version.	Done

Wins, lessons, and next actions

Master Progress Tracker

Use one row per meaningful learning or portfolio session.

[illegible]

Official Learning Resources

Use current official documentation as your source of truth; interfaces and terminology can change.

Resource	Use it for	Visited
Google Analytics demo account	Practise with real business data without affecting a live property.	☐
Google Analytics Academy / Skillshop	Official courses and Google Analytics certification information.	☐
GA4 events overview	Understand the event-based measurement model.	☐
GA4 Explorations	Use advanced analysis techniques beyond standard reports.	☐
GA4 funnel exploration	Build and analyse multi-step journeys.	☐
GA4 path exploration	Analyse event and page/screen sequences.	☐
Campaign URL builders	Create campaign URLs with UTM parameters.	☐
Tag Manager Preview and Debug	Test container changes before publishing.	☐
Tag Manager data layer	Understand how interaction data is passed to tags.	☐
Verify consent mode implementation	Use Tag Assistant to validate consent behaviour.	☐
GA4 BigQuery export setup	Learn how GA4 event data can be exported to BigQuery.	☐
GA4 BigQuery export schema	Study the exported event schema.	☐
Google Analytics Data API	Automate reporting and build custom integrations.	☐
GA4 Measurement Protocol	Understand server-to-server and offline event augmentation.	☐
Looker Studio documentation	Build reports, calculated fields, filters, and controls.	☐

Resource discipline

Documentation supports practice; it does not replace practice. After every learning resource, create an output: a test, query, implementation, diagram, dashboard, or written decision.

Final Completion Gate

Mark the roadmap complete only when you can demonstrate the full analytics workflow.

End-to-end proof

Done	Milestone / Task	Evidence
☐	I can turn a business objective into a KPI tree and tracking plan.	Project
☐	I can implement or specify meaningful events and parameters.	Project
☐	I can validate data before analysing it.	Project
☐	I can analyse acquisition and on-site behaviour.	Project
☐	I can build a funnel or journey analysis.	Project
☐	I can create a dashboard for a named stakeholder.	Project
☐	I can work with exports in spreadsheets.	Project
☐	I can write and validate useful SQL queries.	Project
☐	I can propose and prioritise recommendations.	Project
☐	I can explain privacy, consent, and data risks.	Project
☐	I have at least four polished case studies.	Portfolio
☐	I can present one case study in five minutes without notes.	Interview

ROADMAP COMPLETION DATE

The goal is not to know every menu. The goal is to produce trustworthy evidence that improves a decision.